Linear Algebra (background)

Matrix multiplication:

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{pmatrix}$$

If a matrix A can satisfy the equation $Av = \lambda v$ with some vector v:
v is an **eigenvector** of A.
λ is an **eigenvalue** of A.

Recipe:

$$Av = \lambda v$$

$$(A - \lambda)v = 0$$

$$\begin{pmatrix} a_{11} - \lambda_{11} & a_{12} \\ a_{21} & a_{22} - \lambda_{22} \end{pmatrix} = 0 \implies \det A = 0$$

$$(a_{11} - \lambda)(a_{22} - \lambda) - a_{21}a_{12} = 0 \text{ then solve for } \lambda$$

Dirac notation (background)

We can represent mathematical objects more clearly using Dirac notation.

$$\text{Ket vector: } |\psi\rangle = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \dots \end{pmatrix}$$

$$\text{Bra vector: } \langle\psi| = (\psi_1 \quad \psi_2 \quad \dots)$$

$$\text{Inner product: } \langle\phi||\psi\rangle = (\psi_1 \quad \psi_2 \quad \dots) \begin{pmatrix} \psi_1 \\ \psi_2 \\ \dots \end{pmatrix}$$

Inner product = 1 if and only if $|\phi\rangle = |\psi\rangle$

Inner product = 0 if and only if $|\phi\rangle$ and $|\psi\rangle$ are orthogonal.

Complex numbers (background)

The imaginary unit i is defined by: $i^2 = -1$

Numbers can have a **real part** and an **imaginary part.** These are called complex numbers.

Example 0.1: $3 + 4i$

The complex conjugate of a complex number is the number with the same real part, but the imaginary part has the opposite sign.

Example 0.2: the complex conjugate of $3 + 4i$ is $3 - 4i$.

To find the magnitude of a complex number, multiply the number by its complex conjugate. The answer is the magnitude squared. (so the magnitude is the square root of this value)

Example 0.3: $|3 + 4i|^2 = (3 + 4i)(3 - 4i)$

$= 9 + 12i - 12i - 16i^2 = 9 - 16(-1) = 5$

So the magnitude is $\sqrt{5}$

Euler's rule: $e^{i\phi} = \cos\phi + i\sin\phi$

Classical information (background)

The fundamental unit of classical information is the **bit**.

One single bit contains the answer to one yes/no question.
Yes = 1, No = 0.

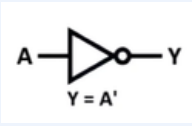
We can combine bits to store more information.
If we have N bits, we can have 2^N possible states.

Example 0.4: If we have a sequence of 4 bits, how many possible states are there?
2^4 = 16 possible states

Example 0.5: Convert 1100 from binary to base-10
Multiply the 0th (right-most) bit by 2^0, the 1st bit by 2^1, the nth bit by 2^n, and add them all together.
$1100 = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$
$= 8 + 4 = 12$

Example 0.6: Convert 13 from base-10 to binary
Break the number into powers of 2 listed in descending order:
$13 = 8 + 4 + 1 = 2^3 + 2^2 + 2^0$
So the 0th, 2nd and 3rd bits are 1, any remaining bits are 0.
The right-most bit is the 0th bit.
(8+4+1) → 1101

Classical gates (background)

Name of gate & what it does	Single or multi bit?	Circuit component diagram
Not - Flips 0 → 1, 1 → 0	Single	
And - Outputs 1 if both input bits are 1, outputs 0 otherwise	Multi	
Or - Outputs 1 if either of input bits are 1	Multi	
XOR - Outputs 1 if strictly one of the input bits is a 1. Outputs 0 if both are 1 or none of the input bits are 1.	Multi	

CHECK YOUR UNDERSTANDING

Question 0.1: Find the eigenvectors and eigenvalues of $\begin{pmatrix} 9 & 8 \\ -6 & -5 \end{pmatrix}$

Question 0.2: convert 49 into binary

Question 0.3: convert 1000011 into base-10